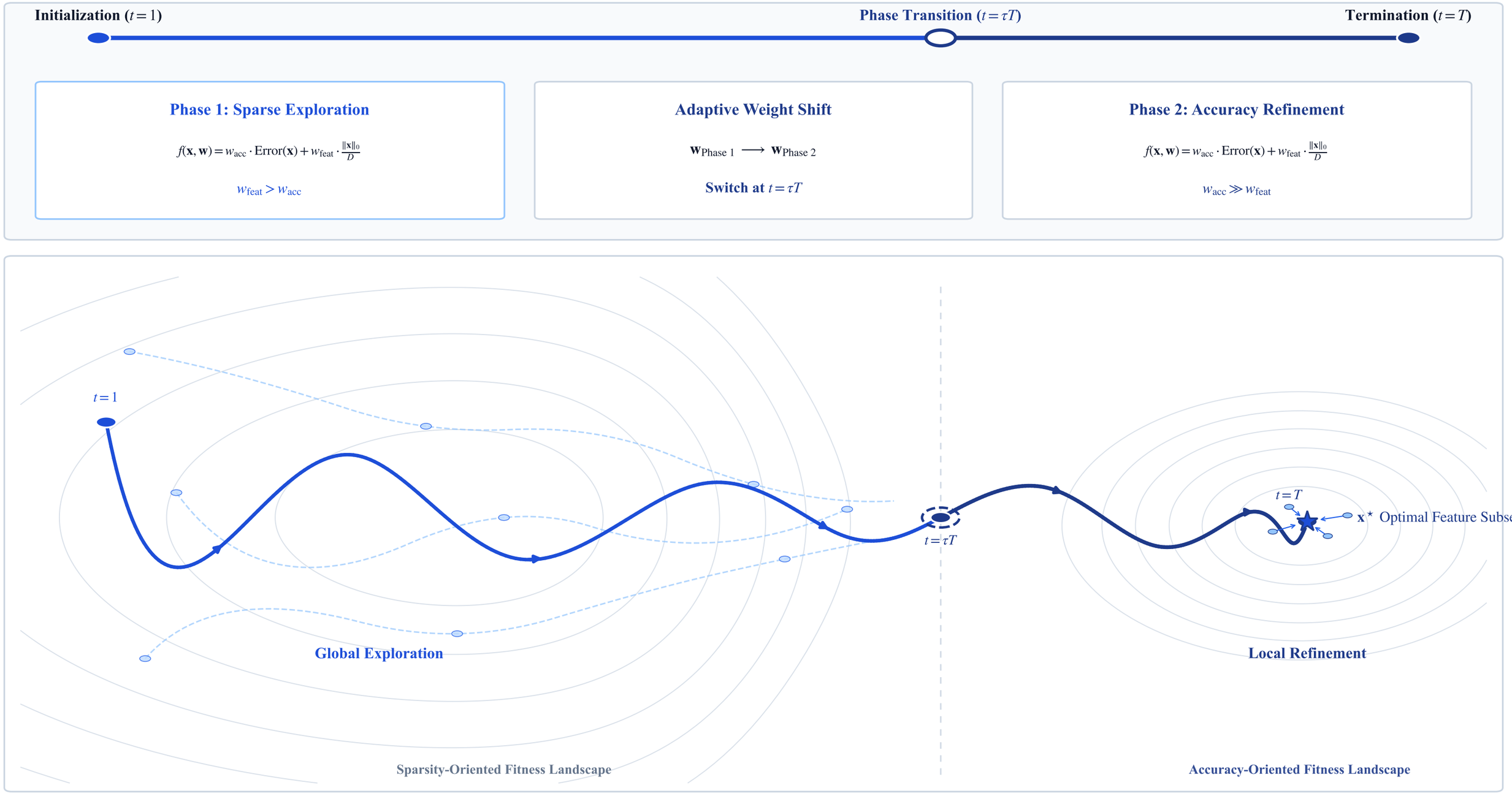

Initialization ($t=1$)
Phase Transition ($t=\tau T$)
Termination ($t=T$)
Phase 1: Sparse Exploration
$f(\mathbf{x},\mathbf{w}) = w_{\text{acc}} \cdot \text{Error}(\mathbf{x}) + w_{\text{feat}} \cdot \frac{\|\mathbf{x}\|_0}{D}$
$w_{\text{feat}} > w_{\text{acc}}$
Adaptive Weight Shift
$\mathbf{w}_{\text{Phase 1}} \longrightarrow \mathbf{w}_{\text{Phase 2}}$
Switch at $t=\tau T$
Phase 2: Accuracy Refinement
$f(\mathbf{x},\mathbf{w}) = w_{\text{acc}} \cdot \text{Error}(\mathbf{x}) + w_{\text{feat}} \cdot \frac{\|\mathbf{x}\|_0}{D}$
$w_{\text{acc}} \gg w_{\text{feat}}$
$t=1$
Global Exploration
$t=\tau T$
$t=T$
$\mathbf{x}^\star$ Optimal Feature Subset
Local Refinement
Sparsity-Oriented Fitness Landscape
Accuracy-Oriented Fitness Landscape